FBG tilt angle prediction

Md Minha Uddin | Roll no. 2401CS39 | Machine Learning Laboratory, Experiment 06 | 13 October 2026

What I did

I used the two experimental sheets from the given workbook. Each has 37 actual readings, including row 1. The theoretical sheet and the extra chart data on the right of the workbook were left out. The target is angle in degrees; the starting inputs are the two wavelength changes (columns D and E).

I tried the original pair (F1), their difference (F2), their average (F3), and difference plus average (F4). I checked the supplied F and G columns against my calculations; they match. The four choices are not four independent sources of information: F1 and F4 are just invertible ways of writing the same two readings.

Validation and leakage control

For each axis I used five GroupKFold outer folds, grouped by angle so repeated measurements at one angle stay together. Within each outer training set, four grouped inner folds selected the feature configuration and model hyperparameter. StandardScaler sits inside each fitted pipeline. I joined the 37 outer-fold predictions before calculating MAE, MSE, RMSE and R-squared. Training time is the mean final fit time per outer fold, excluding the tuning search; it is only a rough comparison on this tiny dataset.

One model from start to finish: GPR

Flow: experimental D/E changes → compute F1/F2/F3/F4 → grouped outer training/test split → grouped inner selection of features and RBF/Matérn kernel → fit scaler on training rows → fit Gaussian process → predict held-out angles → collect errors. Repeat for all five outer folds.

Algorithm: 1) Make five angle groups. 2) For each outer fold, score every allowed feature/kernel pair in four inner grouped folds using RMSE. 3) Refit the best pair on the whole outer training split. 4) Predict the untouched outer test split. 5) Aggregate all 37 predictions. Seed 42 is used for the stochastic tree models.

Dataset checks

Axis	Rows	Angle range	Missing	Exact duplicate rows	Repeated angles
Axial	37	-90 to +90 deg	0	2	17 angle values
Transverse	37	-90 to +90 deg	0	0	17 angle values

The axial sheet has two exact repeats; I kept them because they are recorded observations. Most non-endpoint angles have two readings and zero has three. No rows were added or imputed. The full descriptive statistics and every row-wise prediction are saved with the code.

Data: Kuo Li, *FBG biaxial tilt sensor using one optical fiber*, Mendeley Data, doi:10.17632/f4csv2xjk6.1 (CC BY 4.0).

Feature check (five grouped folds; linear model only)

Axis	F1 pair	F2 diff	F3 average	F4 diff+avg
Axial	6.68	6.17	55.42	6.68
Transverse	46.54	53.89	53.29	46.54

Values are RMSE in degrees. Axial F2 is already strong by itself. For signed transverse angle, even the best linear pair remains weak; average alone mostly describes tilt magnitude.

Nested-tuned model family comparison: axial

Model	MAE	MSE	RMSE	R2	Fit ms	Feature
GPR	1.80	8.6	2.93	0.997	0.81	F1
Polynomial	2.60	18.4	4.29	0.993	0.96	F2
KNN	4.68	37.2	6.10	0.986	0.61	F1
SLR	4.99	38.1	6.17	0.986	0.68	F2
Elastic Net	4.76	40.4	6.36	0.985	0.68	F1
Lasso	5.11	41.7	6.46	0.984	0.69	F2
MLR	5.15	44.7	6.68	0.983	0.68	F1
Ridge	5.06	46.9	6.85	0.982	0.75	F2
Random Forest	6.02	49.2	7.01	0.981	43.18	F1
SVR	4.98	53.3	7.30	0.980	0.74	F1
Gradient Boosting	8.41	82.5	9.08	0.969	20.15	F1
Decision Tree	10.00	100.0	10.00	0.962	0.72	F2

Nested-tuned model family comparison: transverse signed angle

Model	MAE	MSE	RMSE	R2	Fit ms	Feature
GPR	14.95	343.8	18.54	0.870	0.84	F4
KNN	29.10	1324.7	36.40	0.499	0.63	F4
Polynomial	27.33	1377.7	37.12	0.479	0.99	F1
Lasso	39.24	2158.1	46.46	0.184	0.98	F4
Ridge	39.41	2165.2	46.53	0.181	0.74	F4
MLR	39.48	2166.1	46.54	0.181	0.70	F4
Elastic Net	39.15	2194.4	46.84	0.170	0.69	F4
Random Forest	37.82	2903.9	53.89	-0.099	29.48	F4
SLR	48.85	2934.7	54.17	-0.110	0.70	F3
SVR	46.94	3117.9	55.84	-0.180	0.71	F4
Gradient Boosting	42.76	3246.3	56.98	-0.228	20.06	F4
Decision Tree	43.29	3701.9	60.84	-0.401	0.71	F2

All requested hyperparameter configurations

Each row below is the grouped outer-fold score for that fixed setting, with feature choice still inside training. The family comparison on page 2 uses nested hyperparameter selection, so it is the fair final comparison.

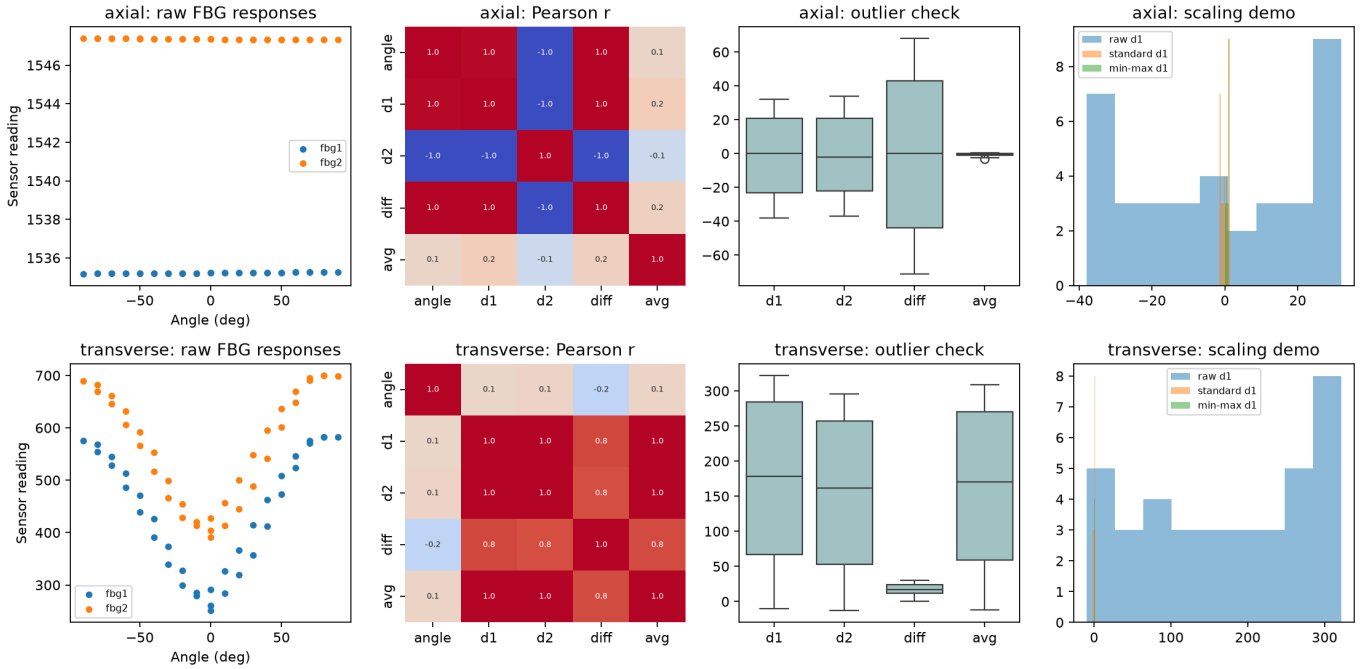
Axial

Model	Configuration	RMSE	R2
SLR	default	6.17	0.986
MLR	default	6.68	0.983
Polynomial	degree=2	5.46	0.989
Polynomial	degree=3	4.29	0.993
Ridge	alpha=0.1	6.85	0.982
Lasso	alpha=0.1	6.46	0.984
Ridge	alpha=1	6.27	0.985
Lasso	alpha=1	6.56	0.984
Ridge	alpha=10	9.90	0.963
Lasso	alpha=10	12.46	0.941
Elastic Net	alpha=0.1, l1_ratio=0.5	6.36	0.985
Elastic Net	alpha=1, l1_ratio=0.5	12.62	0.940
SVR	kernel=linear, C=10, epsilon=0.1	7.30	0.980
SVR	kernel=rbf, C=10, epsilon=0.1	14.09	0.925
Decision Tree	max_depth=2	20.03	0.848
Decision Tree	max_depth=3	13.75	0.928
Decision Tree	max_depth=5	10.00	0.962
Random Forest	n_estimators=50	7.18	0.980
Random Forest	n_estimators=100	7.07	0.981
KNN	n_neighbors=3	6.10	0.986
KNN	n_neighbors=5	7.00	0.981
KNN	n_neighbors=7	9.32	0.967
Gradient Boosting	learning_rate=0.01, n_estimators=100	18.88	0.865
Gradient Boosting	learning_rate=0.1, n_estimators=100	9.08	0.969
GPR	kernel=RBF	3.20	0.996
GPR	kernel=Matern	2.93	0.997

Transverse

Model	Configuration	RMSE	R2
SLR	default	54.17	-0.110
MLR	default	46.54	0.181
Polynomial	degree=2	37.90	0.457
Polynomial	degree=3	38.88	0.428
Ridge	alpha=0.1	46.53	0.181
Lasso	alpha=0.1	46.46	0.184
Ridge	alpha=1	46.67	0.176
Lasso	alpha=1	46.65	0.177
Ridge	alpha=10	49.63	0.068
Lasso	alpha=10	52.58	-0.046
Elastic Net	alpha=0.1, l1_ratio=0.5	46.84	0.170
Elastic Net	alpha=1, l1_ratio=0.5	50.56	0.033
SVR	kernel=linear, C=10, epsilon=0.1	63.94	-0.547
SVR	kernel=rbf, C=10, epsilon=0.1	54.86	-0.138
Decision Tree	max_depth=2	56.05	-0.189
Decision Tree	max_depth=3	62.41	-0.474
Decision Tree	max_depth=5	64.17	-0.558
Random Forest	n_estimators=50	53.97	-0.102
Random Forest	n_estimators=100	45.71	0.210
KNN	n_neighbors=3	31.77	0.618
KNN	n_neighbors=5	44.12	0.263
KNN	n_neighbors=7	49.51	0.073
Gradient Boosting	learning_rate=0.01, n_estimators=100	56.93	-0.226
Gradient Boosting	learning_rate=0.1, n_estimators=100	58.78	-0.307
GPR	kernel=RBF	20.96	0.834
GPR	kernel=Matern	17.00	0.891

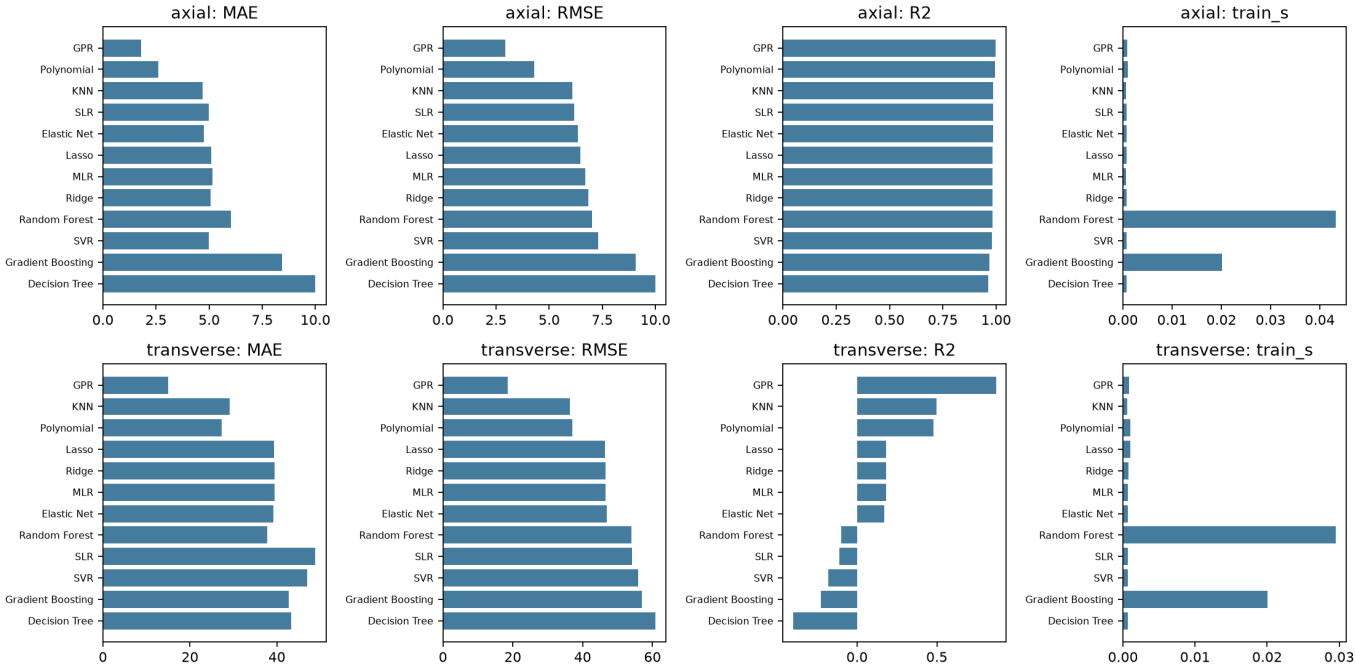
Exploration, sensor response and scaling



The scatter panels plot both FBG readings against angle; the heatmaps show Pearson correlations; boxplots flag extreme wavelength changes. The final panel shows raw, StandardScaler and MinMaxScaler distributions for one input as a visual demo. In actual validation, scalers were fitted only on training folds.

The axial difference follows signed angle much more clearly than the average. Transverse FBG readings rise with absolute tilt in both directions, so a low signed Pearson correlation does not mean the sensor has no tilt information. Boxplot whiskers are a screening aid, not a reason to drop end angles.

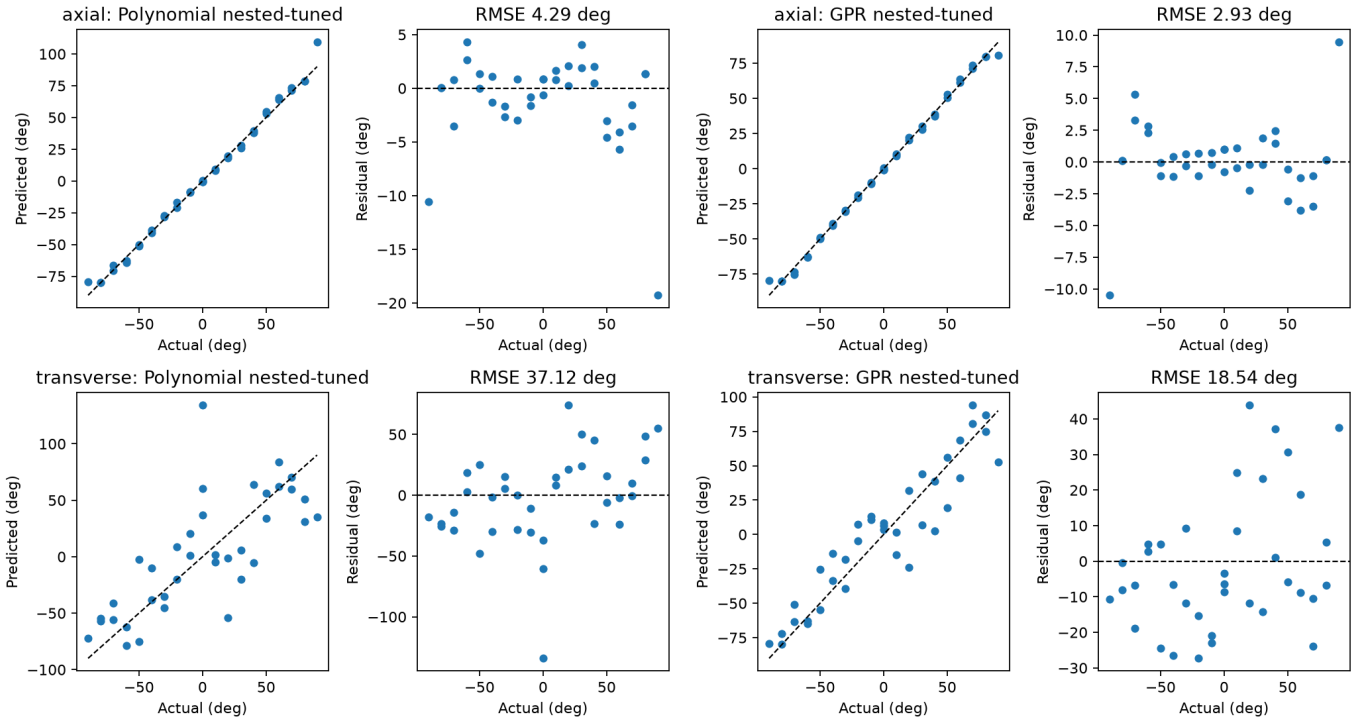
Model-wise accuracy and training time



The four columns show MAE, RMSE, R-squared and mean outer-fold fit time for each axis. These are the nested-tuned family results from page 2. Smaller MAE/RMSE is better; larger R-squared is better. Fit times are too small and environment-dependent to claim a hardware benchmark.

GPR is strongest on both axes in this run. Degree 3 polynomial improves axial accuracy over degree 2, but the extra flexibility does not reliably solve the transverse sign ambiguity. Greater tree depth and more forest trees do not automatically improve grouped prediction.

Actual predictions and residuals



The best statistical family is Polynomial for axial (4.29 deg RMSE) and Polynomial for signed transverse (37.12 deg). The best conventional ML family is GPR for axial (2.93 deg) and GPR for transverse (18.54 deg). These plots use only outer test predictions.

Advanced task C: signed versus absolute transverse tilt

Using the same grouped folds and a simple linear model on average wavelength change, signed angle gives RMSE 53.29 deg (R2 -0.074), while absolute angle gives RMSE 6.75 deg (R2 0.937). That gap is the main thing I learned: the average signal tells me how far the sensor tilted, but generally not which way. Small direction clues in both channels let GPR do better, though I would not assume they stay stable on a new sensor.

My conclusion

For accurate calibration on these 37 readings, I would pick Matérn GPR for axial tilt and also test it for signed transverse tilt. For a simple real-time monitor, a linear differential model is easier to explain and fast for axial angle. I would report transverse magnitude from the average unless direction has been separately calibrated and checked on more runs. The 37-row dataset is tiny, so these errors are evidence for this experiment, not a guarantee in deployment.

Public code and full row-wise results: https://github.com/mdminhaj-2106/ML_Lab9.